



Fig. 4: Contactless payment jacket by Barclaycard and Lyle & Scott (Image courtesy: Barclaycard)

out doubt, healthcare is one of the most valuable applications of smart clothing, and several products are emerging in this space, ranging from simple bedsheets that monitor health parameters of the patient, to super innovative gait trainers. Ekso Bionics has special wearable bionic suits that help in neuro-rehabilitation, for victims of stroke or spinal cord injury. The suits help patients to walk again, through gait-training exercises.

Combining a bit of fashion with utility, fashion designer Pauline van Dongen has developed a knitted cardigan, which has integrated stretch sensors that measure the movements of elderly persons wearing the cardigan, and communicate this data to the service provider. This cardigan is comfortable enough to be used as daily clothing. In fact, van Dongen specialises in such tech-powered fashion. She has also developed garments that help correct posture, as well as water-resistant jackets with flexible solar panels that help the wearers charge their devices when outdoors.

An intelligent t-shirt developed by scientists in Madrid can locate patients within a hospital using a GPS system that works in closed spaces. It also finds out whether the patient is seated, lying down, walking or running. This is very useful to monitor people with memory loss problems.

SensFloor goes one step further—the tech does not even have to be worn. The textile is embedded with

sensors that measure capacitance, and is designed as an underlay for carpets. Basically, by measuring capacitance, sensors can detect when someone walks, sits or lies on the carpet. Radio modules help the sensors transfer collected data to a control module, which analyses the activity.

SensFloor can be assembled at healthcare centres or at assisted care accommodations. It lets patients carry out simple activities by themselves, without being bound to their beds. At the same time, it helps to keep the caregivers aware of the patients' movements. The system can be adapted to act when certain conditions are observed. For example, it can alert the patient with lights or beeps when there is the risk of collision with walls or furniture, or when he or she is too close to the staircase. It can also be made to alert caregivers. The smartcarpet underlay can also be used for other purposes like home care and energy conservation.

This is just the tip of the iceberg when it comes to smart fabrics for healthcare.

A dash of fun with smart fabrics

Applications of smart fabrics are not all serious. There is a major fun element to it, too. Wearable Experiments (WeeX), for example, have developed a product called Alert Shirt for Australian television group Foxtel, to give viewers of a football game the physical sensation of being part of the game. They can choose a player, whose movements they wish to experience. The football players wear a smartjersey, which is equipped with sensors that capture various physical aspects such as heartbeat, impact with the ball, collision with other players, falls and so on. Sensor data is sent to the base station and broadcast along with the video footage.

Viewers, in turn, wear jerseys that have several tiny motors to replicate the physical sensations felt by the players. The jerseys are



SINCE 1962

PROGRAMMABLE DIGITAL PANEL METERS



SNMP635SS



SNMP35SS

Features

- TRMS Measurement
- Easy Programming
- 4 Digit Super Bright Red LED Display
- High Accuracy
- Setup / Programming Protected by Password
- Dual Aux. 110V AC/230V AC
- 1A / 5A AC Input in the same Meter (User Selectable) for Ammeter
- 50-500V AC Input for Voltmeter
- User Programmable Display from 0-9999 as per the CTP / PTR
- Auto Selection of Decimal Point

An ISO 9001:2008 Certified Company

MECO INSTRUMENTS PRIVATE LTD.

Plot No. EL-1, MIDC Electronic Zone,
TTC Industrial Area, Mahape,
Navi Mumbai - 400710 (INDIA)

Tel : 0091-22-2767 3300 (Board), 2767 3311-16 (Sales)
Fax : 0091-22-27673310, 27673330

Email : sales@meconist.com
Web : www.meconist.com